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RESEARCH ARTICLE

A Machine Learning-Based Surrogate Finite Element Model for Estimating Dynamic Response of Mechanical Systems

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ABSTRACT An efficient approach for improving the predictive understanding of dynamic mechanical system variability is developed in this work. The approach requires low model assessment time through the fitting of surrogate models. ML-based surrogate algorithms for finite element analysis (FEA) are developed in this study to accelerate FEA and prevent rerunning complex simulations. The research begins with an overview of the recent novelties in ML algorithms applied to finite element (FE) and other physics-based computational schemes. To predict the time-varying response variables, that is, the displacement of a two-dimensional truss structure, a surrogate FE technique based on ML algorithms is developed. In this work, several ML regression algorithms, including decision trees (DTs) and deep neural networks, are developed, and their efficacies are compared. In this study, the ML-based surrogate FE models are able to effectively predict the response of the truss structure in two dimensions over the entire structure. Extreme gradient-boosting DTs provide more precise outcomes and outperform other ML algorithms.

INDEX TERMS Surrogate modeling, finite element analysis, mechanical system analysis, machine learning, artificial neural networks, random forest trees, gradient boosting regression trees, adaptive boosting trees.

I. INTRODUCTION

Machine learning (ML) approaches in science and engineering have gained rapid popularity in recent times. In the framework of computational engineering predictions, this implies the development of efficient, inexpensive, rapid, and reliable ML prediction methods without compromising the complexity of the high/mid-fidelity finite element (FE) models [1]. Transient vibration analysis is a vital part of any mechanical and system design and health condition evaluation. To achieve the best results in structural health monitoring (SHM), the orientation of sensor/actuators, as well as the frequency and domain of the propagated wave for each structure, is critical [1]. These characteristics are typically discovered using numerous FE models, which is both time-consuming and costly. The method presented in

this study has the potential to reduce these expenditures significantly.

Therefore, the present study explores the feasibility of developing ML surrogates for predicting the dynamic response of a mechanical system. The most popular approach to modeling these mechanical systems involves the utilization of mid-fidelity physics-based models inserted into various structural solvers. This approach can be computationally expensive if the system is complex. A low-cost alternative will involve using surrogates to estimate response variables of mechanical systems, owing to their potential for leveraging the black-box nature of the problem and appealing computational simplicity [1]. ML models are particularly promising in scientific and engineering challenges associated with partially understood processes [2], [3], [4]. These models can also be employed in cases where the running of numerical or analytical models at desired resolutions in space and time is computationally infeasible. ML algorithms have been employed in combination with finite element

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analysis (FEA) in preliminary attempts, particularly those aimed at addressing static mechanical systems. However, the efficiency of regression ML algorithms in forecasting displacements in time-varying truss structure systems has rarely been assessed. The present study aims to apply and validate ML algorithms in order to predict time-varying displacement as a surrogate for FEA in a two-dimensional framework (truss structure) and then compare the efficiencies of these algorithms. Displacement is correlated with the strain of a truss structure, and hence every unexpected response of the structure that impacts the strain can affect the displacement of the structure. Changes in displacement, whether abrupt or gradual, might cause structural damage including fractures and plastic deformation [1]. Moreover, displacement is considered a serviceability design feature since extreme displacements are undesirable for users and may damage surrounding sensitive structures [1].

In this research, we dealt with under-fitting and over-fitting issues of the constructed ML models since the model tried to predict fourteen nodal displacements in 2 directions (twenty-eight outputs) by having the external excitation loadings in one direction. Hence, various DNN structures with different numbers of hidden layers and units, activation functions, and gradient optimization methods, and also, a variety of gradient boosting and bagging tree structures, have been developed in this research to solve the problem. However, higher dimensions (more than one dimension) mechanical systems may need a tremendously large training dataset due to the model's increased complexity. The constructed models achieved acceptable results as surrogate models using smaller training datasets and observations in comparison with previously conducted research.

II. BACKGROUND

Owing to recent developments in ML and artificial intelligence (AI), ML techniques could be considered a promising solution for generating data-driven surrogate models that can accurately predict desired quantities. Various applications of ML techniques, including engineering, design, bioengineering, material science, medicine, and technology, require a large amount of data [5]. Supervised ML algorithms approximate the relationship between certain input variables and numeric outputs using various optimization algorithms that minimize the estimation error. An error function specifies the distance between an output value in a dataset and the related output obtained from the simulated model. The error function and optimization technique for each ML algorithm is unique. Training a model involves a process of optimization over existing data. ML has recently been used in computational mechanics to formulate multiscale elements [6], enhance the performance of traditional elements [7], and create data-driven-based solutions [8]. Capuano et al. [9] used ML techniques to obtain a multiscale FE algorithm with low computational cost. They employed ML to derive a direct relationship between the element states (outputs) and their forces (inputs) and avoid the

complex task of finding the internal displacement field and removing numerical iterations. An approximated model (surrogate) of the elements is produced by incorporating the FE of interest into an ML algorithm. Surrogate modeling of fluid dynamic systems using ML has also been attempted in a few cases. Annabestani et al. [10] combined microfluidic impedance flow cytometers with ML to introduce a set of sheath-free microchannels. They estimated the size and number of multiparticle transiting channels. In mechanical behavior prediction of structures, Tan et al. [11] focused on the development of a real-time prediction model by coupling the spatiotemporal correlation with external load through an autoencoder network based on SHM data. An autoencoder mechanism was performed to obtain the high-level representation of raw monitoring data at different spatial positions, and the time series temporal correlation was deciphered using the recurrent neural network.

In structural mechanics, ML can predict vibration and dynamic responses using measured parameters during operations based on high-fidelity FE models to tradeoff high accuracy and low computational costs. The application of ML techniques combined with FE methods has recently attracted research attention. Thus, improving the tradeoff between computational accuracy and computational costs is essential. Using supervised ML algorithms as FE surrogates have been considered recently in biomechanical engineering, e.g., in the stress analysis of the aorta [12], [13], and human liver [14]. In computational mechanics, ML has recently been used to formulate multiscale elements [9], to improve the performance of conventional elements [15], and create data-driven solvers [15], [16], [17]. The ability to accurately estimate the actual dynamic response of a mechanical system during usage is pivotal to defining accurate maintenance procedures and refining the design procedures of modified complex systems [12]. However, in general, several extensive complex FE simulations are required during the lifespan of a mechanical system, which is often impractical. Low-/mid-fidelity FEAs, which are updated on the actual condition of the system via experimental measurements and update themselves rapidly and effectively, have been offered as alternatives to comprehensive simulations [18], [19]. Bartilson et al. [20] explored natural frequency and mode shape sensitivities in structures with at least one plane of symmetry and distinct natural frequencies. Jang and Smith [21] used a full-scale FE model in a simulation-based study to investigate the effects of temperature distribution on natural frequencies. Castro-Toscano et al. [22] presented a technical vision system for using ML regression algorithms in structural behavior analysis. Hashemi et al. [23] proposed a unique semi-analytical approach as a surrogate for FE modeling of a wind turbine blade where the lateral deflection of the structure is derived. FE simulations could also primarily train an ML model and prevent rerunning complex simulations. Najib and Nobari [24] introduced a procedure based on nonlinear

autoregressive with an exogenous input model as an alternative shortcut method for extracting the damage indicators. They used the finite element model of a beam connected to a rigid support via a flexible adhesive layer to study the efficiency of the proposed method. Hall et al. [25] presented a new technique for developing both linear and nonlinear statistical forecasts of the winter North Atlantic Oscillation (NAO) based on complex systems modeling, that their polynomial NARMAX models demonstrated considerable skill in out-of-sample forecasts and their performance was superior to that of linear models. Zhu et al. [26] introduced a data-driven surrogate model, known as the Nonlinear in Parameter AutoRegressive with exogenous input (NP-ARX) model, to circumvent the difficulties in the analysis and design of fan systems. Ren et al. [27] proposed two active learning approaches to combine Kriging and ANN models for reliability analysis the efficiency and accuracy of the proposed approaches were demonstrated by representative examples and two finite element problems. To address the challenging problem of unsatisfactory accuracy and the high sensitivity of surrogate prediction in the presence of uncertainties, an uncertainty-based design optimization framework based on the proposed multi-fidelity polynomial chaos-Kriging (MF PC-Kriging) surrogate model was proposed by Zhao et al. [28], with particular superiority for complex aerodynamic applications. Wang et al. [29] proposed a multi-objective optimization method based on the Kriging surrogate model and finite element analysis to mitigate the radial recoil and foreshortening ratio of bioresorbable magnesium alloy stent and investigate the impact of strut thickness on stent expansion behavior. Swischuk et al. [30] considered parametric surrogate models for applications in science and engineering where the goal was to predict high-dimensional output pressure. The proposed methodology develops a low-dimensional parameterization of pressure using the proper orthogonal decomposition (POD) and combines this parametrization with ML methods to learn the map between the input parameters and POD expansion coefficients [30]. The authors used two engineering cases to investigate the relative costs and efficacy of four ML techniques: neural networks (NNs), multivariate polynomial regression, k-nearest neighbors, and decision trees (DTs). Chu et al. [31] discussed the significance of scaffolding collapses, explored a method that performs scaffolding monitoring, and established an integrated method; this research cross-connects various components (e.g., strain data, FE model-based structural analysis, ML, and an actual scaffold) in the presented framework.

The ML algorithm that discriminates between scaffolding situations, such as safe, overturning, uneven-settlement, and over-loading, uses load-related strain data as training and testing sets. Paul et al. [32] have designed and developed essential components of a scientific framework for developing a data-driven model-based real-time control system that employs FE methods for solving time-dependent heat equations and developing the database. Oh et al. [33] presented

a sustainable strain sensing model using an artificial NN (ANN) to estimate the strain responses of columns depending on the wind-induced behavior of high-rise buildings. Their ANN model is based on evolutionary learning, comprising training in a radial basis function (RBF) NN and evolving in a genetic algorithm. Karballaezadeh et al. [34] used three ML methods-Gaussian process regression, M5P model tree, and random forest (RF)-to predict structural numbers in flexible pavements. The employed dataset is related to 759 flexible pavement sections in the Semnan and Khuzestan provinces in Iran. It includes “structural number” as output and “surface deflections and surface temperature” as inputs. Liu et al. [35] proposed solutions based on ML models, such as regression trees (RTs) and NNs. Their feasibility and value are demonstrated through the analysis of fracture toughness measurements. They found that solutions based on RTs and NNs could provide accurate results for a specific problem, but NN-based solutions outperformed RT-based solutions in simplicity. Using well-known incremental-iterative solution techniques is a computationally expensive method of efficiently designing geometrically nonlinear structures. Therefore, Mai et al. [36] developed a deep NN (DNN)-based surrogate model integrated with a differential evolution (DE) algorithm to solve the optimum design problem of a geometrically nonlinear space truss under displacement constraints. They referred to the approach as DNN-DE. Deng [37] proposed a method for extracting the stress distribution characteristics of indoor load-bearing buildings to improve the safety analysis using a hybrid method. This method used a combination of a genetic algorithm and ML to extract the stress analysis features of the buildings. Truong and Pham [38] employed the support vector machine (SVM) algorithm to develop a metamodel for predicting the ultimate strength of trusses using direct analysis. This analysis considered several kernel functions for the SVM model, including linear, sigmoid, polynomial, and RBF. They used a planar 39-bar nonlinear inelastic steel truss to investigate the performance of the kernel functions. Sakaguchi et al. [39] presented a method for extracting features of nearly optimal brace types and locations for large-scale steel building frames subjected to static seismic loads. Using SVM with an RBF kernel, a new formulation was provided for extracting key characteristics of brace kinds and positions from ML outcomes. Kim et al. [40] employed a high-fidelity FE method and ML solutions to determine the accurate strengths of steel circular hollow section X-joints with minimal computational effort. Amani et al. [41] studied analytical functions for estimating the temperature-dependent behaviors of poorly and highly dispersed graphene oxide-reinforced nanocomposite materials in the framework of an ML-based approach. Rahman et al. [42] presented a data-driven approach to estimate the shear strength of steel-reinforced fiber concrete (SFRC) beams. They incorporated a large database compilation of 507 experimental data points comprising the shear span ratio to effective depth, concrete compressive

strength (CCS), longitudinal reinforcement ratio, volume fraction, aspect ratio, and type of fiber. In the study, 11 ML models were tested for shear strength estimation of SFRC beams: linear regression, ridge regression, lasso regression, DT, RF, SVM, k-nearest neighbors, ANN, extreme gradient boosting (XGBoost), adaptive boosting (AdaBoost), and CatBoost. Zhang et al. [43] leveraged the recent advances in DL and developed a physics-guided CNN (PhyCNN) for data-driven structural seismic response modeling. The aim was to develop deep PhyCNN models for structural response prediction based on limited seismic input-output datasets (either from simulation or sensing) and physics constraints. Subsequently, the created surrogate model was used for fragility analysis based on specific limit state criteria. Furthermore, an unsupervised learning method based on K-means clustering was suggested for splitting datasets into training, validation, and prediction categories, allowing the most efficient use of limited datasets. Fan et al. [44] proposed a novel dynamic response reconstruction approach for SHM using densely connected CNNs, employing skip connection and dense block techniques in the designed network architecture. This approach facilitated information flow and increased the training efficiency and accuracy of feature extraction and propagation with fewer parameters in the network (compared with the number required for other approaches). Subpixel shuffling and dropout methods were employed in the proposed network to minimize computational load and increase training efficiency. A supervised manner was used to train the network. The measurements of the available channels at response available locations and desired channels at response unavailable locations were the input and output for the applied supervised method, respectively. Zhang et al. [45] conducted a study to develop advanced DL approaches for nonlinear structural response modeling and prediction. The authors proposed two long short-term memory (LSTM) network schemes for data-driven structural seismic response modeling. Unlike the classical physics-based nonlinear time history analysis, which uses numerical approaches, the suggested DL model, trained on existing datasets, could precisely predict both elastic and inelastic responses of building structures in a data-driven manner. Furthermore, the seismic inputs were clustered using an unsupervised ML algorithm based on a proposed dynamic K-means clustering approach to (1) produce the smallest but most insightful datasets for training the LSTM and (2) enhance the prediction precision and robustness of the model trained with limited data.

Azim and Gul [46] introduced a damage detection framework for element-level damage identification for railroad truss bridges. This framework combines the analysis of acceleration with that of strain responses. A sensor-clustering-based time series analysis approach and damage characteristics were studied in terms of structural nodes from the truss bridge employed to analyze the acceleration response. The strain data, which provided information on damage from instrumented truss components, was evaluated

via principal component analysis. Rather than using a strain gauge for SHM, Wagner et al. [47] tried to measure strain through two-dimensional electrical impedance tomography combined with a planar elastoresistive sensor. Shamshirsaz et al. [48] developed a smart method for monitoring structure integrity utilizing lamb wave propagation and signal processing systems. Mokhtari et al. [49] employed piezoelectric nanogenerators to absorb ambient-induced vibrations of structures and convert these vibrations to a signal that can be analyzed for SHM. Barkhordari and Eshaghi [50] evaluated the ability of regression models and a hybrid technique of ANN and simulated annealing (SA)-ANN-SA to predict responses of reinforced concrete shear walls subjected to strong ground motions. Predictions, including maximum acceleration, maximum velocity, and earthquake characteristics, were employed in the study using various ML techniques. Furthermore, the accuracies of the models in recognizing the responses of the shear walls were compared. Asteris et al. [51] tried implementing a hybrid ensemble surrogate ML technique in predicting CCS—an important parameter for durability design and service life estimation of concrete structures in civil engineering projects. A database comprising 1,030 records was employed to train and validate four conventional ML models—ANN, linear and nonlinear multivariate adaptive regression splines, Gaussian process regression (GPR), and minimax probability machine regression. The expected outputs of the conventional ML models were merged and trained using ANN to create a hybrid ensemble model. Mianroodi et al. [52] proposed a DNN model as a fast surrogate model for local stress calculations in inhomogeneous nonlinear materials. The DNN model could recreate the stress distribution in geometries that were different from those used for training. The trained model simulated the micromechanics with a mean absolute percentage error of 6.4% in a single forward network assessment without iteration. This was achieved for elastoplastic materials with up to four times mechanical contrast in yield stress across adjacent regions. Vaishali et al. [53] presented a stochastic dynamic characterization of functionally graded shells using an efficient SVM-assisted FE approach. A Monte Carlo simulation was combined with an ML-based FE computational framework to obtain a comprehensive probabilistic description of natural frequencies. Individual and compound impacts of depth-wise source uncertainty in material properties of functionally graded shells were considered in the stochastic approach. The influences of various critical parameters, such as the power-law exponent, temperature, thickness, and variation of shell geometries, were investigated. Badarinath et al. [54] proposed a surrogate FE approach based on ML algorithms to estimate the TV response of a one-dimensional beam. They employed many ML regression models, including ANNs and DTs, to estimate the stress distribution, and their performances were compared for direct estimation of the stress distribution over a beam structure. Zhu et al. [55] introduced a data-driven surrogate model, referred to as the

nonlinear in parameter autoregressive with exogenous input (NP-ARX) model, to circumvent the difficulties associated with the analysis and design of fan systems. NP-ARX is a linear input-output model that obtains the nonlinear model coefficients of the design parameters of interest. The frequency response function is useful in system analysis and design. Le [56] developed a surrogate GPR-based ML model to predict the axial load of square concrete-filled steel tubular (CFST) columns under compression. An experimental database collected for this purpose was used in developing and training the GPR model. The model outperformed existing models in terms of the axial load of square CFST columns. Kneifl et al. [57] implemented and evaluated a nonlinear nonintrusive model reduction approach for structural dynamic models. The temporal evolution of reduced system states was learned based on combined classical projection approaches and ML regression algorithms. Physics-guided ML is a new paradigm that blends data-driven and physics-based ML models to exploit ML data discovery capabilities without losing significant physics/domain knowledge. Yu et al. [58] proposed a novel physics-guided ML method based on re-current NN and multilayer perceptron for structural dynamics simulations. The core idea was to integrate the underlying physics of structural dynamics into a data-enabled guide in the training phase of ML models. Zheng et al. [59] proposed a novel design approach based on an iterative ML algorithm to accelerate the topological design exploration of compression-only shell structures with planar faces, considering structural performance and construction constraints. They explained that the construction of NNs helps one train a surrogate model to evaluate the structural performance of many potential frame structures without a considerably slower geometric form-finding procedure (than those associated with models without NNs). Using the big-data-driven ML approaches, Raj et al. [60] developed a DL-based algorithm known as UNet to predict the local strain fields in a two-phase composite material subjected to a uniaxial tensile load. To reduce the computational cost of determining stress distributions in heterogeneous media using FEA, Feng and Prabhakar [61] employed DL to develop the first-ever difference-based NN frameworks based on engineering and statistics knowledge. They focused on discontinuous domains that manifest high-stress concentrations. In the past, various FEA model geometries and stresses were directly employed as inputs for training a NN. However, the authors emphasized differences in stress distribution across distinct input samples to improve the prediction accuracy in heterogeneous media.

III. ML-BASED DYNAMIC RESPONSE ESTIMATION METHODOLOGY

Dynamic response estimation of different structures during operations can be considered a significant part of any mechanical and system design and health condition evaluation. This paper presents a unique FEA-based ML approach that directly predicts the dynamical response

of two-dimensional mechanical systems. In this section, the concept of the methodology and target structure is presented.

A. METHODOLOGY

The approach proposed in this research will combine FE models and ML regression algorithms to develop an ML-based surrogate model for FEA. A mid-fidelity surrogate model is constructed by utilizing supervised ML techniques. FEA simulations provide needed displacement data for training ML models to estimate displacement in real-time based on further measurements. The algorithm will concentrate on predicting overall strains since the displacement of every targeted node on a two-dimensional truss structure is directly related to the strain. In truss structures, the strain and relative displacement are directly related, and hence the relative displacement can be obtained by transforming the measured strain values [62]. In general, two types of methodologies may be used to determine displacement: direct measurement methods (contact and non-contact) and indirect measurement methods. On real structures, indirect displacement measurement utilizing sensors such as accelerometers [63], tilt sensors/inclinometers [64], or strain sensors [65] might be advantageous. Strain sensors transduce external mechanical stimuli into electrical signals. The advantage of using a network of strain sensors to determine the movement of a structure is that the same sensor network can also be used to assess other monitoring parameters. These parameters include the neutral axis [66], prestressing forces [67], thermal identity verification [68], and curvature [69]. Therefore, the present study focuses on a strain-based method. This method is very versatile and may be employed to train and predict other mechanical response variables such as stress, velocity, and acceleration. The FE model is fed the actual condition of a time-dependent mechanical system by measuring the response (i.e., strain in this study). Afterward, the ML algorithm estimates the output of the system at any other nodes. As the first step of the proposed approach, a FE model of the system of interest is constructed using ANSYS code. We build a suitably big training dataset that reflects the expected loading circumstances by performing a series of time-varying FEA simulations. To derive the response associated with each degree of freedom and its derivatives ($\gamma_1, \dot{\gamma}_1, \ddot{\gamma}_1 \dots, \gamma_i, \dot{\gamma}_i, \ddot{\gamma}_i$), the model is loaded with several loading scenarios ($L_1, \dots, L_i$).

$$\begin{aligned} L_1 &\Rightarrow \{\gamma_1, \dot{\gamma}_1, \ddot{\gamma}_1\} \\ L_2 &\Rightarrow \{\gamma_2, \dot{\gamma}_2, \ddot{\gamma}_2\} \\ &\vdots \\ L_i &\Rightarrow \{\gamma_i, \dot{\gamma}_i, \ddot{\gamma}_i\} \end{aligned} \quad (1)$$

The loading conditions can be imposed sequentially to reflect continuous external load variations.

$$[L_1, \dots, L_i] \longrightarrow [\gamma_1, \dots, \gamma_i], [\dot{\gamma}_1, \dots, \dot{\gamma}_i], [\ddot{\gamma}_1, \dots, \ddot{\gamma}_i] \quad (2)$$

A fourteen number of nodal locations are considered in the entire system. The desired values at these locations are defined as $n_1(), \dots, n_m()$ where element m indicates the total number of nodal locations. In the subsequent step, the input and output of FEA simulations are set as input and output variables for the ML algorithm. The ML approach creates a correlation between the desired numerical response, i.e., strain in the nodal locations, and external excitation loading that is applied to the structure at each time step. After constructing ML models, the desired numerical responses at all locations are predicted based on the external excitation loads which will be applied to the structure. The proposed method can be organized as an ML-based surrogate model for deriving the considered response variables as illustrated in Fig 1.

B. TARGET STRUCTURE

The analysis focuses on two-dimensional flexible, dynamic systems with known mechanical characteristics such as material properties and geometry. A two-dimensional truss structure is considered. Fig 2 depicts a 31-bar planar truss structure with simple support, which is modeled using 31 classical dependable finite elements without internal nodes, thereby resulting in 25 degrees-of-freedom [70]. The truss elements are 1.52 m long and 0.05 m wide. The material density and elasticity modulus are 2770 kg/m³ and 70 GPa, respectively, and the cross-sectional area of the bar is 0.0025 m².

C. FE MODEL OF THE STRUCTURE

In this work, the FE model of the desired structure is constructed in ANSYS code in order to obtain desired responses of the truss structure and generate needed data. A hexahedron mesh (size of mesh: 0.05 m) is used based on the geometry of the desired structure in the FE modeling. The boundary conditions are assumed as simply supported and roller-hinged for two tips of the structure.

IV. DATA GENERATION

The FE model of the target structure is developed using ANSYS code. A surrogate FE model is developed based on a numerical environment where a FE solver generates both training and prediction data. The numerical setting prevents any experimental errors and noise impact on the predicted results. The FE model is used to obtain strain along the entire structure. The strain at nodal locations, external excitation loadings, and time operation is used in model development. In addition, the target structure is a 31-bar planar truss structure with simple support, which is modeled without internal nodes, resulting in 25 degrees-of-freedom [70]. The target structure, selected nodes, and structural elements are shown in Fig 2. To excite the entire structure, a sinusoidal distributed external load (frequency range: 5 Hz $\times$ 40 Hz) is applied to element number 31 of the structure in the X direction. The time response is numerically integrated using a

time step of $\Delta t = 1 \times 10^{-4}$ s to prevent aliasing. The duration of the simulation was 0.3 s.

A. DATA PREPARATION

The ML models are built using the data collected by FE models in this study. The dataset includes over 1.5 million data points. The dataset consists of 16 attributes: time instant and applied external loading, and the strain $s_1, s_2, s_3, \dots, s_{14}$. Each strain indicates the mean magnitude of strain recorded in the time step at different nodes on the truss structure when an external force is applied to the structure. The aim is to determine the strains in the 14 considered points using applied external excitation across a set of time instances. All attributes have a numerical datatype, and no values are deemed outliers because the experimental data is developed using the FE model. The applied external excitations and related time are employed as input attributes, and the 14-point strains comprise the output variable to be anticipated. Z scores normalization technique is used for the data since the range of attribute values is unusually wide in both X and Y directions. In ML modeling, the normalization approach is a popular method of scaling data. This transfer is achieved by subtracting the value of each variable from the overall average of the value (v_i) and dividing by the overall standard deviation of that variable (σ) as follows:

$$Z_i = \frac{v_i - \mu}{\sigma} \quad (3)$$

Through this technique, the original distribution of data is transferred to a Normal Z-Distribution.

V. MACHINE LEARNING MODELS DEVELOPMENT

This study employs four ML models for regression, including DNNs and three DTs models, in order to forecast strain based on strains previously prescribed throughout certain periods of time and in specific locations over the truss structure. DTs are easy to build and understand and can handle both nominal and continuous inputs [71]. Therefore, this predictive analytic approach is widely used by practitioners. XGBoost trees, AdaBoost trees, and RF are some of the DTs commonly used for classification and regression [72].

XGBoost, AdaBoost, RF, and DNNs are employed in the present study. Each algorithm and the best model structure for each technique are briefly described in the next section. In addition, the implementation details of the research are provided. Through the GoogleColab platform [54], the algorithms are developed with Python libraries, including SKlearn, Keras, and XG-Boost (extreme gradient boosting).

A. DEEP NEURAL NETWORKS

ANNs are referred to as non-linear statistical data models that replicate the role of biological neural networks [73]. ANNs are increasingly attractive, effective, efficient, and successful in achieving pattern recognition in many problems [74]. DNNs were employed during the development of the predictive models and can be represented as a series of layers,

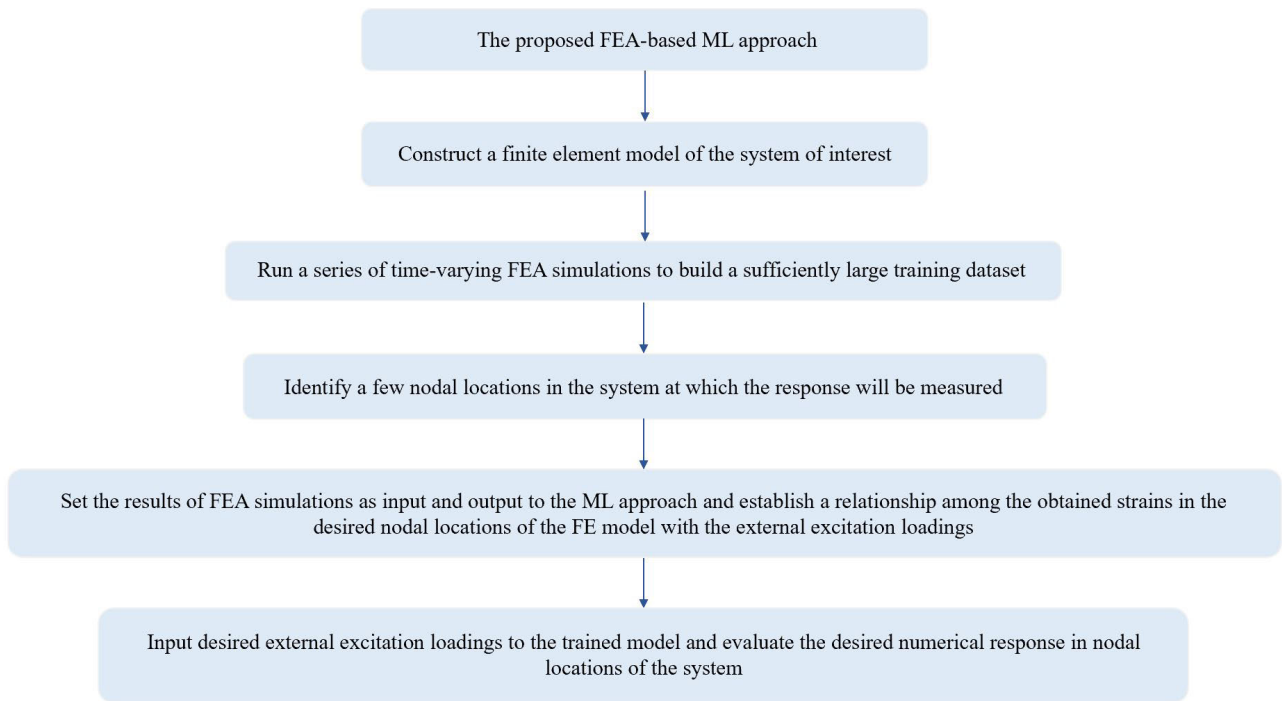


FIGURE 1. Flow chart of ML training approach.

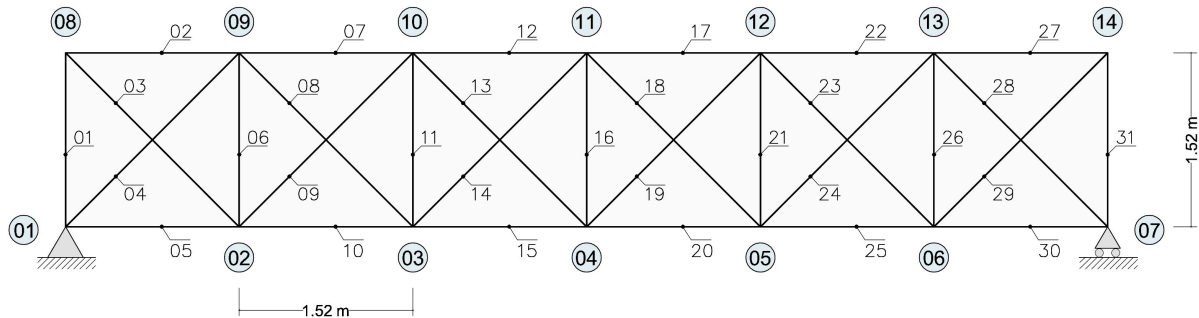


FIGURE 2. A 31-bar plane truss structure.

which are described by:

$$f(x) = f_L(x) \circ \dots \circ f_1(x), \quad (4)$$

where every layer applies a linear transformation, and then an element-wise nonlinearity. Numerous aggregating layers increase the predictive power of the model. The output signal of every unit in a layer of a multilayer NN can be computed using the output signal of the previous layer [1]. After the weighted sum is computed, an activation function is used to generate the output for the subsequent layer. Activation functions define the output of a node associated with a set of inputs, and hence the selection of such functions is important [1]. Commonly used types of DNNs include logistic (sigmoid), hyperbolic tangent (Tanh), Softplus, and rectified linear units (ReLU). ReLU networks were utilized in this work. ReLU was selected based on its compatibility

with every sampling method, as the data distribution has no effect on the training process of the network [75]. The number of hidden layers, type of activation function, number of nodes, and learning rate of the algorithm should all be considered during the creation of DNNs [76]. The overall structure of DNNs is composed of the input, hidden, and output layers (Fig 3). Two key architectural features must be considered during the development process of DNNs: the number of layers and the number of hidden units to choose for each layer. Compared with fewer hidden units, a greater number of hidden units (a higher-dimensional representation space) helps the network to learn more complex representations but increases the computational cost of the network. Moreover, undesirable patterns may be learned (patterns that will improve performance on the training data (without improving performance for the test data) and cause

TABLE 1. Details for DNNs structure in this research.

Layer	Structure Criteria	ReLU Structure
Input Layer	Factors	strains
	Number of Units	2
Hidden Layer	Number of Hidden Layers	9
	Number of Units	2016
Output Layer	Activation Function	ReLU
	Dependent Variables	strain
	Number of Units	14
	Error Function	Mean Square Error

overfitting). Backpropagation, a form of gradient descent used to compute the gradient considering the weights in a NN, is often utilized in multilayered network training. Multiple methods for optimizing the gradient descent approach have been recently developed. In the backpropagation process, gradient de-scent methods are required for maximizing the network performance by altering the weights. We employed the following five gradient descent optimization algorithms in the present study.

1) ADAGARD

Dynamically incorporates information about the geometry of the data observed in previous iterations to perform informative gradient-based learning [77].

2) SGD

A stochastic approximation for lowering an objective function (a sum of differentiable functions).

3) ADADELTA

A per-dimension learning rate method that dynamically adapts over time using only first-order data [78].

4) RMSprop

Divides the learning rate of a weight by a running average of recent gradient magnitudes associated with that weight [79].

5) ADAM

Only requires first-order gradients with low memory requirements and calculates individual adaptive learning rates for various parameters [80].

The best performance was achieved with the Adam algorithm for optimizing the performance of the network, considering the dataset and the regression challenges.

In this work, the best performance among all constructed DNNs is realized for one input layer with two units, nine hidden layers, and one output layer with 14 units. The model was trained for 500 epochs, with ReLu activation function, and a learning rate of 0.001, using the Adam optimizer in Python. Table 1 provides further details of the constructed DNNs.

B. EXTREME GRADIENT BOOSTING TREES

XGBoost DTs, a prominent data mining approach that has recently been developed by Chen and Guestrin [81], are based on a combination of classification and RTs of a training process. XGBoost, an extended version of the

ensemble learning method (gradient boosted DT), features an enhanced loss function and loss optimization processes [82]. This method can be employed for classification with a categorical output variable and regression for estimating a numerical target [83], [84]. The main purpose of the XGBoost algorithm is to create a classification RT using all data attributes. Every classification and RT is adjusted to the errors during the training phase, indicating the discrepancy between the observed and predicted values of the output variables from the ML model [1]. To realize this adjustment and minimize errors, the tree is compared with the previously constructed tree. Furthermore, compared with many other ML algorithms, XGBoost is computationally efficient, performs well in finding the correlation among input and output variables [85], avoids overfitting, and requires minimum feature engineering. The lack of model generalization is a significant challenge in modeling dynamic systems such as the system considered in this study. In other words, the ML model may have a high accuracy rate on the training data, but its performance degrades when applied to the test data. XGBoost models have demonstrated greater generalization (than the ML model), indicating that they improve model accuracy on test data [86]. Consequently, we employed the XGBoost technique as one of the most effective DTs for regression modeling. The variables for the trees and related boosting should be considered to achieve minimum prediction error because XGBoost has applications for regression and classification targets [87] (Table 2). We employed the GridSearch option for selecting the optimal number of parameters for the DTs, based on the dataset required for obtaining an ideal number of parameters. For regression considered in this study, the highest predictive performance for XGBoost was obtained for 1000 trees with gbtree. A learning rate of 0.08 and a maximum depth of 11 (which is the optimal number of splits in each tree node) with subsamples of 1 and col samples of 1 were employed. The L1 regularization was employed only in ridge regression and was, therefore, zero, whereas the L2 regularization value was 1. The loss function is linear regression that evaluates the sum of the squares of the model error while predicting future output values in relation to the seen values in the dataset.

C. ADAPTIVE BOOSTING TREES

The Ada Boost algorithm, short for adaptive boosting, is a boosting approach that is used as an ensemble method in ML. The most suited and therefore most common algorithms used with Ada Boost are DTs. Boosting is a concept that originated in the field of supervised learning, which is the automated learning of an algorithm that is based on labeled data with seen outcomes. The aim is to generate correct predictions for unlabeled future and unseen data. Boosting helps in reducing variance and bias in an ML ensemble. Supervised learning is a type of ML that also includes unsupervised learning using unlabeled data and semi-supervised learning, which combines supervised and unsupervised learning [88].

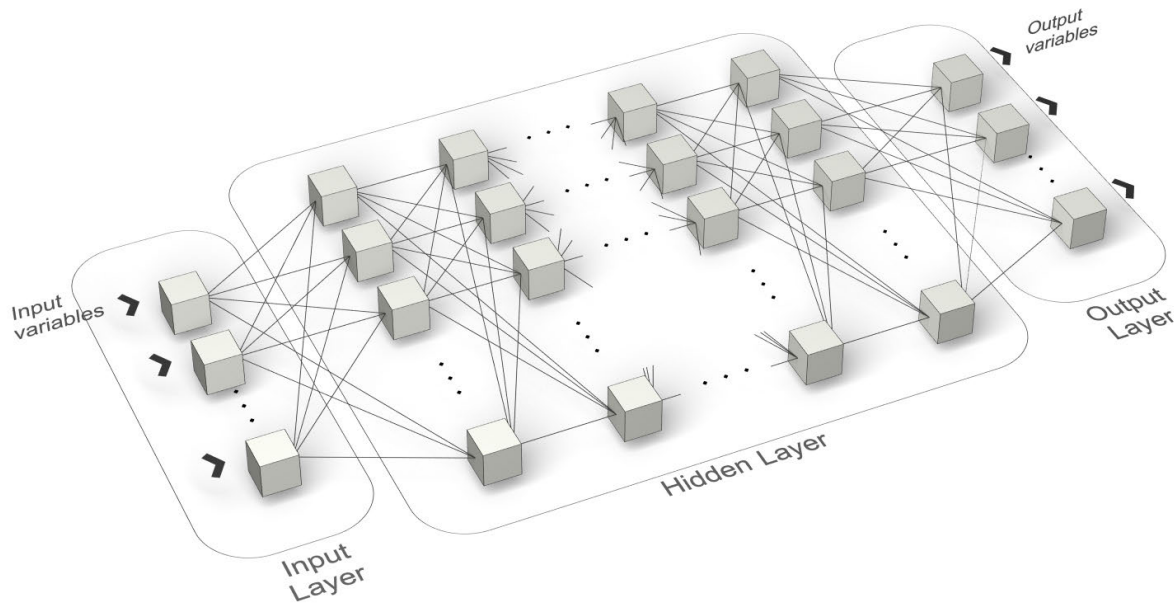


FIGURE 3. The structure of deep neural networks (DNNs).

TABLE 2. XGBoost (extreme gradient boosting) structure by grid search.

Parameters	Description	Range in Grid Search
Booster	Boosting algorithm used	['gblinear', 'fbtree']
N-Estimator	Number of boosted trees	[900,1000,1200,1500]
Max-depth	Maximum tree depth for base learner	[9,11,13,15]
Subsample	Subsample ration of the training instances	[0.5,1]
Colsample	Subsample ratio of columns for each tree	[0.5,1]
Reg-alpha	L1 regularization used for Ridge regression	[0,0.5,1]
Reg-lambda	L2 regularization term used to reduce high dimension	[0,0.5,1]
Learning-rate	Boosting learning rate	[0.05,0.08,0.1]

TABLE 3. AdaBoost (Ada Boost regression trees) structure by grid search.

Parameters	Description	Range in Grid Search
Booster	Boosting algorithm used	['gblinear', 'fbtree']
N-Estimator	Number of boosted trees	[500,700,1000,1200]
Max-depth	Maximum tree depth for base learner	[11,13,15]
Learning-rate	Boosting learning rate	[0.01,0.05,0.1]

TABLE 4. Random forest regression trees structure by grid search.

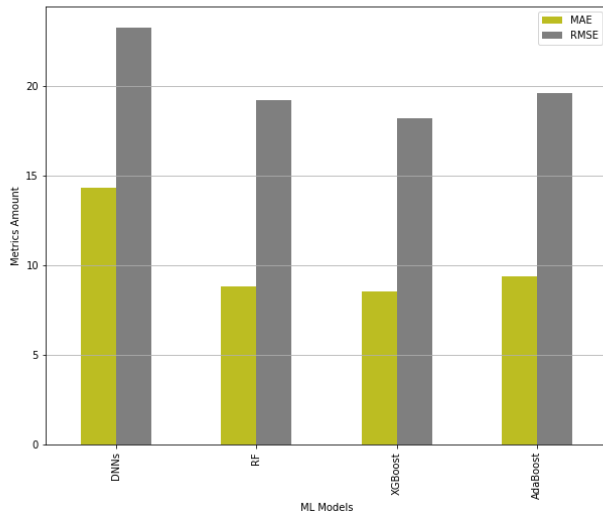
Parameters	Description	Range in Grid Search
N-Estimator	Number of boosted trees	[70,100,120,150]
Max-depth	Maximum tree depth for base learner	[10,20,30,40,50]

Theoretically, boosting was used to determine whether a weak learning tool for classification and regression can be converted into a strong learner [89]. In general statistical models, boosting techniques are commonly utilized for the estimation of unknown quantities. An Ada Boost regressor is a meta-estimator that fits a regressor on the original dataset and then adapts further copies of the regressor to the same dataset. In adapting these copies, the weights of instances are modified in accordance with the error of the current prediction [90], [91]. In Ada Boost, weak

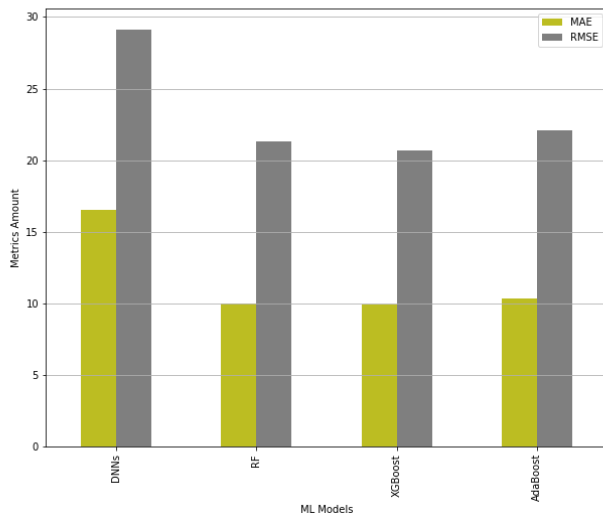
learner predictions are compounded through a weighted accumulation to reproduce the final prediction [92]. Multiple weights, $w_1, w_2, \dots, w_N$, are added to the training set in each boosting iteration to enhance the training data. The first boosting iteration involves these weights and the original training data. The learner algorithm is then reapplied to newly weighted data after each initial weight is adjusted. In the subsequent iterations, the weights of incorrectly predicted training data from the previous step will be raised and the weights of the correctly predicted data will be lowered. Each

TABLE 5. Model comparison metric.

Metric	Formula
MAE	$\frac{1}{n} \sum_{i=1}^N y_i - \hat{y}_i $
RMSE	$\sqrt{\frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{n}}$
R ²	$1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2}$



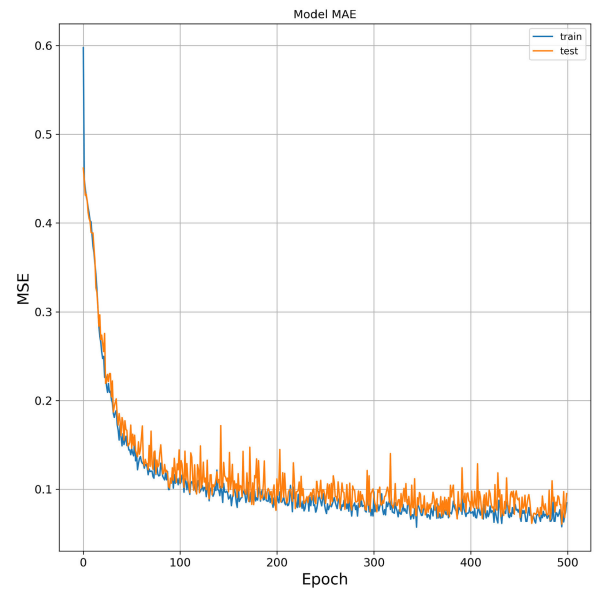
(a) In X direction.



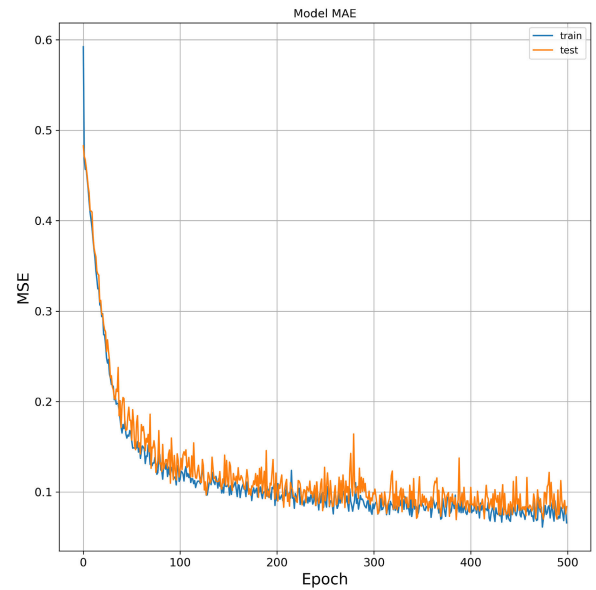
(b) In Y direction.

FIGURE 4. RMSE and MAE of the built models.

consecutive weak learner is then compelled to concentrate on the samples missed in the sequence by the previous learners [93]. Compared with other algorithms, Ada Boost is less prone to overfitting as the input parameters are separately optimized [94]. As a result, the Ada Boost method is employed in this work as one of the most acceptable DTs for regression modeling. The parameters for the trees



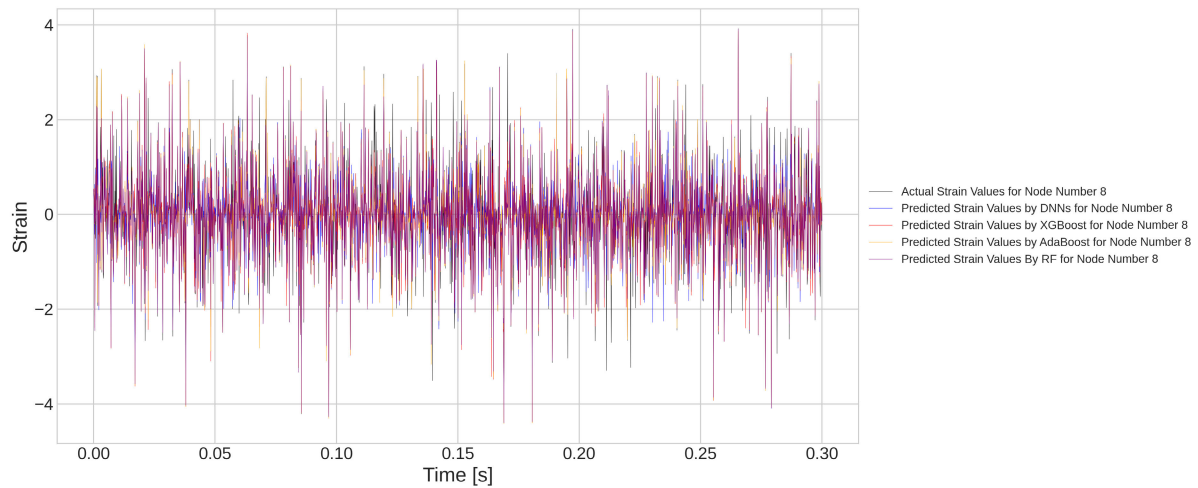
(a) In X direction.



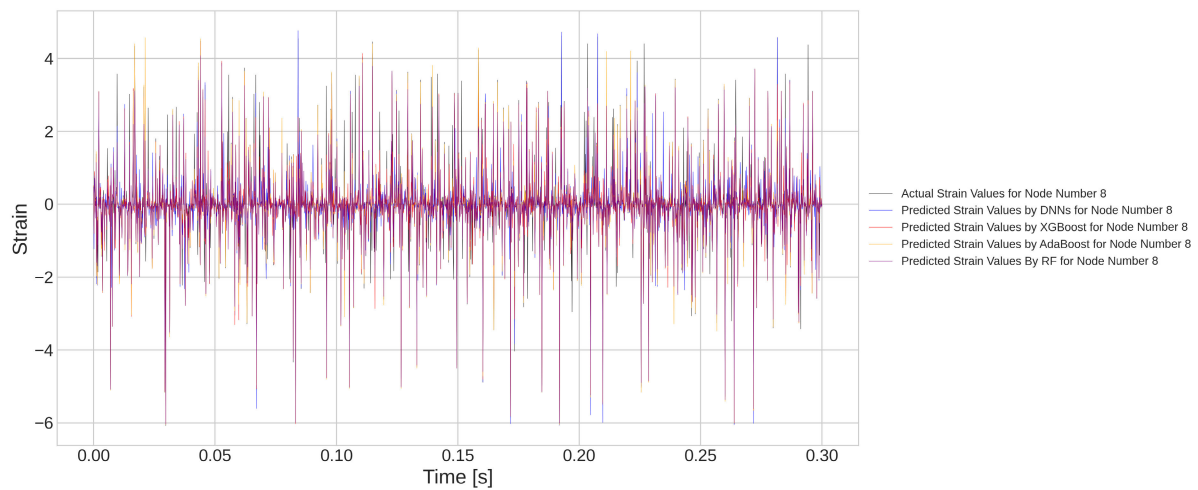
(b) In Y direction.

FIGURE 5. DNNs mean squared error in forecasting the strain based on epoch number.

and the corresponding boosting are listed in Table 3. The GridSearch option for the DTs associated with the dataset is used to achieve the optimal number of boosted trees and the maximum tree depth for the base learner in the Ada Boost regression method. Selected values for the AdaBoost DT with the highest predictive power were 1000 trees for regression, with a learning rate of 0.01 and a maximum depth of 13 (which is the optimal number of splits in each tree node). A minimum sample leaf of 1 and a minimum sample split of 2 were employed. The loss function is linear regression that calculates the sum of the squares of the model error in predicting further output values based on observed values in the dataset [1].



(a) In X direction.



(b) In Y direction.

FIGURE 6. Actual and predicted strain values with time at node number 8.

D. RANDOM FORESTS

RF is a nonparametric ML approach for multiclass classification and regression problems [95]. The RF is a well-recognized and hugely successful bagging type of decision tree ensemble. Individual trees are constructed using bootstrap samples rather than the original sample. This strategy, referred to as bootstrap aggregating or simply bagging, reduces overfitting. In the first step of the process, separate classification and RTs are built during the training phase, with a forecasted estimation of the output variables. The trees are then combined in the second step to obtain an average value for the output variable. To reduce the variance and forecasting bias, the bagging approach can be utilized for developing an RF model. Bagging significantly reduces the variance and prediction bias of the output target for algorithms that have high variance. Moreover, RFs are ideal for nonlinear data [96] and they often perform outstandingly in prediction

tasks. However, one drawback of RFs is that the ease of comprehension associated with individual DTs is lost when numerous DTs are aggregated in the RFs [97].

The number of trees to grow is an important parameter of the RF algorithm. The parameters that are considered for the trees and the corresponding boosting are presented in Table 4. We employed the GridSearch option for selecting the ideal number of RF trees in each forest and the depth of each tree based on the dataset. In addition, we developed RF DTs using 150 trees in each forest; the depth of each tree was 20, and the maximum future was set on auto. A DT regressor is used as the base method for splitting tree nodes, with a mean square error calculated for the loss function.

E. MODEL PERFORMANCE METRICS

Model assessment is aimed at estimating the generalization error of a selected model. That is, the performance of

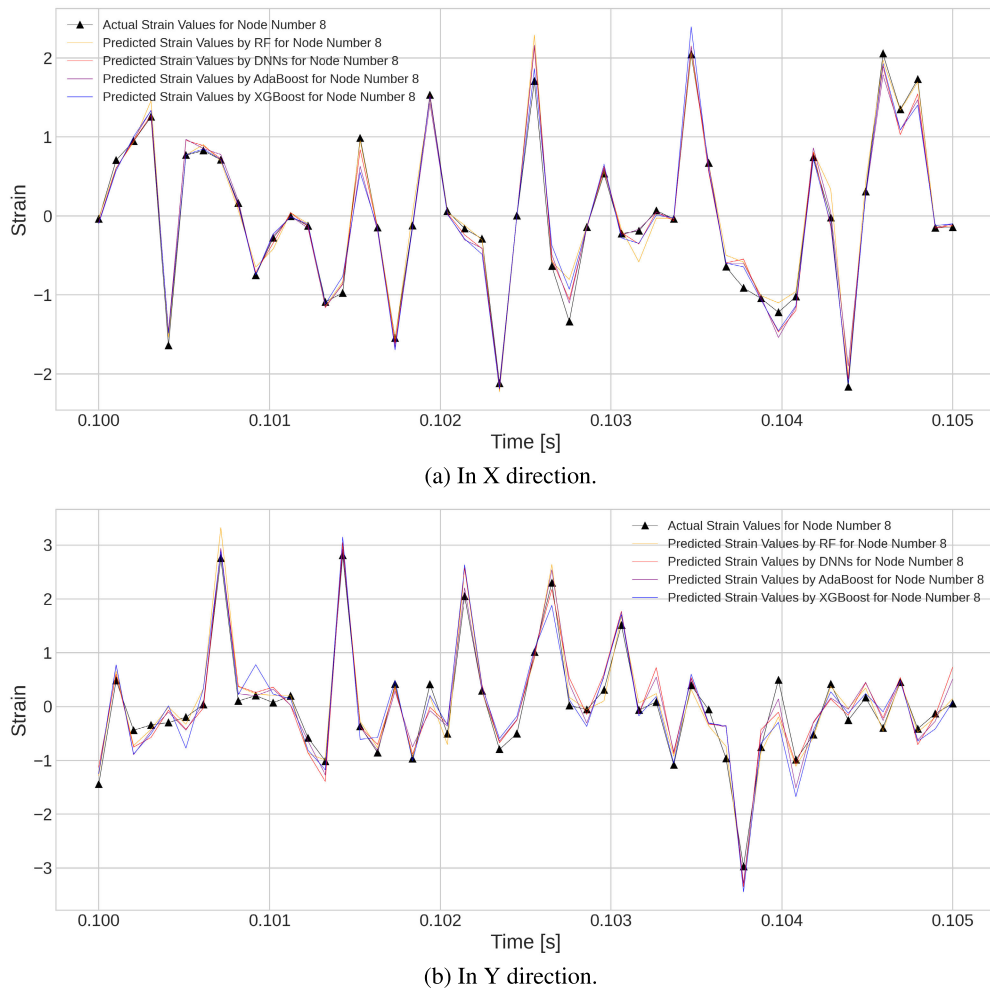


FIGURE 7. Actual and predicted strain values, zoom of time story at node number 8.

the model on data seen during training (else an ML model would memorize the training data) and on unseen data is assessed [98]. Mean absolute error (MAE), root mean squared error (RMSE), and R-squared (R^2) are the quantitative performance indicators for regression modeling. The average difference between the actual values of the output variable in the original data and the anticipated output values of the developed ML models is represented by the MAE. An ideal model exhibits lower MAE (than a non-ideal model), which indicates a more accurate performance of the model in forecasting further occurrences of the output. The standard deviation of the response variables is defined as the RMSE, a parameter for calculating the error of a model in predicting quantitative data. In a regression model, R-squared (R^2 , also known as the coefficient of determination) is a statistical metric that determines the proportion of variation in the dependent variable that can be attributed to the independent variable. To put it another way, R^2 , which ranges from 0 (poor fit, no benefit derived from employing the model over fixed background response rates) to 1 (perfect fit), represents the goodness of the data fit the regression

model. In order to compare models, the model with the highest R^2 and the lowest RMSE and MAE should be chosen. Details regarding the calculation of MAE, RMSE, and R^2 are provided in Table 5.

VI. RESULTS

The outcomes of XGBoost, AdaBoost, RF, and DNNs are presented, and their performance as a surrogate for a FEA are estimated for the two-dimensional mechanical systems employed in this research. A training dataset consisting of 1.22 million data points, accounting for 80% of the total data collected, was used to train four models.

The three DT models have high values of R^2 over 0.98, but the DNN model exhibited lower performance, as evidenced by an R^2 value of 0.92 in both directions. This signifies that the ML algorithms including XGBoost, AdaBoost, and RF can describe over 98% of data variances and are strong predictive models in both X and Y directions. Although, compared with the DT models, the DNNs model achieved lower R^2 values. This trend is different from those observed for the MAE and RMSE in the X direction. The lowest (2.84%

TABLE 6. Model performance in X direction on test data.

Models	R ²	MAE	RMSE
DNNs	0.9270	14.31	23.27
RF	0.9403	8.80	19.22
XGBoost	0.9417	8.55	18.23
AdaBoost	0.9390	9.30	19.61

TABLE 7. Model performance in Y direction on test data.

Models	R ²	MAE	RMSE
DNNs	0.9049	18.45	29.13
RF	0.9422	10.02	21.28
XGBoost	0.9420	9.89	20.54
AdaBoost	0.9391	10.29	22.01

and 3.45%), second lowest values (3.50% and 7.96%), second highest (3.97% and 8.72%), and highest (14.64% and 23.08%) values are observed for the XGBoost, RF, AdaBoost, and DNN models, respectively. In the Y direction, the XGBoost model exhibited the best performance with MAE and RMSE values of 2.94% and 3.57%, respectively. RF exhibits the second-best performance (MAE: 4.02% and RMSE: 8.83%). AdaBoost (MAE: 4.10% and RMSE: 9.05%) and DNNs (MAE: 16.13% and RMSE: 25.37%) exhibit the third-best and poorest performances, respectively. The findings imply that ML DTs models developed for predicting strain in the truss structure are promising. The XGBoost model is the most predictive model, which exhibits excellent performance in both the X and Y directions, considering the fact that the external loading used for the excitation of the desired structure was applied in the X direction.

Next, the trained models are applied to a test dataset containing 307,000 data points which are new for constructed models in order to further assess the performance of the ML models. The practicality of the models is assessed based on their performance on these data when they are applied to a two-dimensional mechanical system (data were not utilized in the training phase and are new to the models). Tables 6 and 7 show the test data findings obtained for the X and Y directions for the ML models.

The values in Tables 6 and 7 indicate the difference between the actual strain (output variable) in the study and the estimated strain in the X and Y directions, respectively, based on the developed ML models. The MAE values indicate the average absolute difference between the strains in the data, and the projected strains using XGBoost, RF, AdaBoost, and DNNs. These strains are 8.57%, 8.80%, 9.30%, and 14.31%, respectively, in the X direction, and are 9.89%, 10.02%, 10.29%, and 18.54%, respectively, in the Y direction. Therefore, XGBoost provided the highest accuracy strain prediction in both directions and DNNs yielded the lowest accuracy. Fig 4 shows the MAE and RMSE graphs for ML models in the X and Y directions, respectively. Fig 5 shows the DNN performance, which is described by the mean squared error. The performance stability is unacceptable for test data up to 500 epochs.

The predicted R² indicates how well a regression model predicts responses for new observations. In this case, the XGBoost model exhibits the best performance with R² values of 0.9417 and 0.9422 in the X and Y directions, respectively. Based on the achieved results, three DT models provide an accurate and precise prediction of the strain with respect to the MAE and RMSE on average. The forecasting performance of the model decreases in the Y direction since the MAE and RMSE of all four ML models (RF, XGBoost, AdaBoost, and DNNs) increased in this direction. This may have resulted from the fact that external loadings to the desired structure were applied along the X direction.

For a physical understanding of the ML forecasting performance, observations of the predicted strain variation along the whole structure and with respect to the time are needed. Four snapshots of the strain variation in the X and Y directions (with different time zoom at node number 8) are depicted in Figs 6 and 7. The strain along the whole structure at a time instant of 0.2015 s is shown in Fig 8. From the results, all three DT models exhibit acceptable overall performance as FEA surrogates, thereby validating the statistical indexes. The XGBoost and RF algorithms provide better predictions of strain (than those provided by the other algorithms) over the whole structure at all nodal locations. Considering the type of data employed and the goal of this effort, ensemble modeling techniques including bagging and boosting perform satisfactorily. This is true, although (compared with the other algorithms) the deep learning algorithm appears to have more difficulties in anticipating dynamic responses in both directions.

The results provided in Tables 6 and 7 confirm that the ML models that are correctly developed, could be considered successful surrogate models for an FEA. That is, these models exhibit acceptable accuracy and precision in estimating strain in a two-dimensional framework at a given location based on prior external excitation forces at a time instant. However, the DNN model could not completely support these results. XGBoost is a new and widely used algorithm that is successfully employed in solving many regression problems. RF and AdaBoost are considered two of the most powerful DTs in regression applications. The study found that, when dealing with dynamic truss structure data, DNNs are unable to create this correlation and predict the strain with the same high precision and low error rates achieved for the XGBoost and RF models. In general, the achieved results indicate that the ML approach is feasible for two-dimensional applications, and can be used confidently as a surrogate for FEA. The constructed ML models (notably the XGBoost and RF models) can successfully estimate strain values at every location based on the magnitude of external excitations applied to the desired structure. More complex applications (than two-dimensional applications), such as examining three-dimensional cases, may benefit from this concept; however, a larger training dataset would be needed for higher-dimensional cases, owing to the additional complexity of the model. The fundamental problem in using

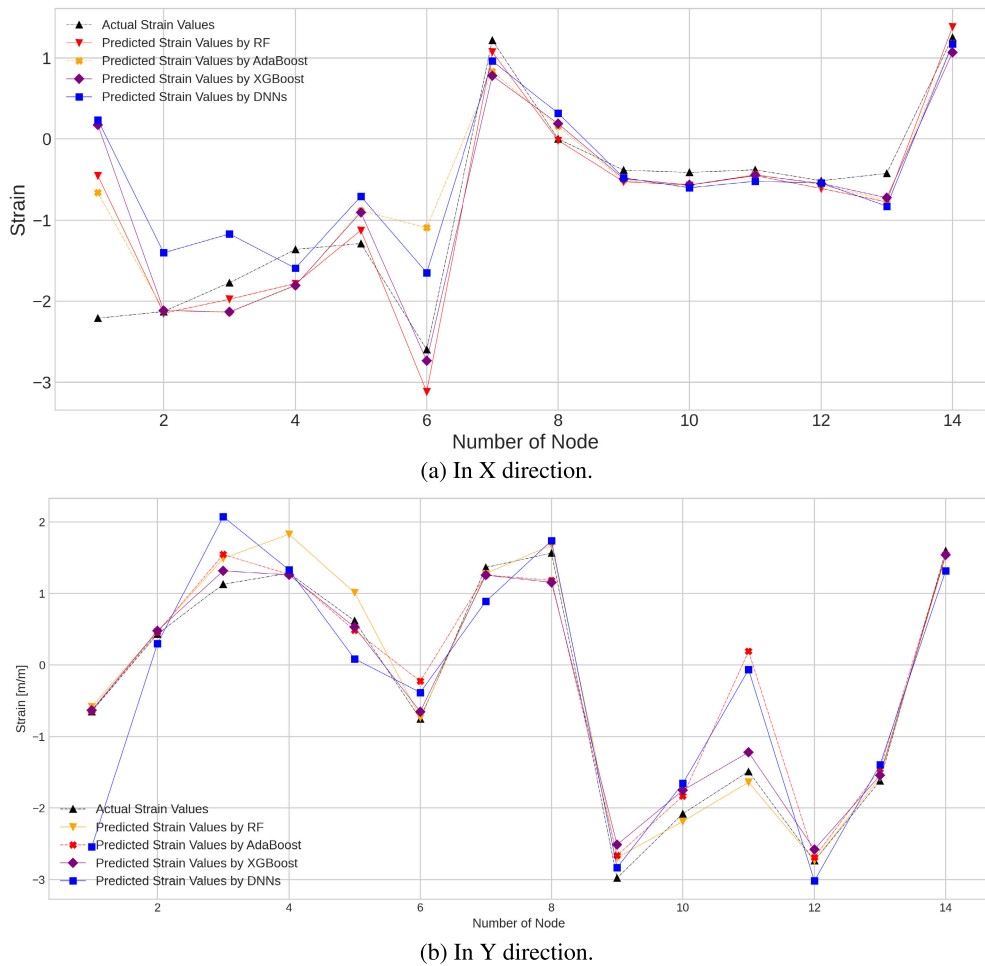


FIGURE 8. Actual and predicted strain values at $t = 0.2015$ s.

an ML model as a surrogate for FEA lies in the type of data and instability in the values over time, and the huge training sets required; the datasets must contain millions of data records and must include the expected loading conditions that the dynamic system would experience. However, acceptable results were obtained in this piece of work by using a smaller dataset than expected.

The main reason for employing ML models is to improve the trade-off between computational accuracy and computational costs since using numerical and physics-based models is tremendously costly for studying the dynamic behavior of mechanical systems with a large number of DOF. The constructed ML models showed significant performance for the desired structure in two dimensions. In the literature, many ML models were used in numeric values prediction. For supervised ML modelling in this piece of work, various ML regression algorithms with different learning mechanisms were employed and their initial efficacy was evaluated. Top four ML models that showed sensible results were selected and then, varied structures were constructed to achieve significant results. The details of final built ML models were briefly described in the previous section.

Furthermore, additional analyses (as part of future studies) are necessary for developing and employing more appropriate ML techniques (than those utilized here) to obtain more accurate results and determine the best placements of the virtual sensor locations. In the future, authors will study more complex structures for achieving a better perception of using ML algorithms to investigate dynamic mechanical systems. In addition, as an advantage of employing ML algorithms over other competing methods, the ML models can be used as a pipeline for real-time operation. The analysis of results shed light on the difficulties and possibilities of developing an ML pipeline as a surrogate for an FEA that has numerous applications for response variable prediction of various structures in other industries.

VII. CONCLUSION

A unique machine learning-based surrogate model for estimating the dynamic response of a time-varying two-dimensional mechanical system is proposed in this study. The main contribution of this work is a novel approach that combines ML techniques and a FE approach in order to describe transient systems. A two-dimensional 31-bar planar

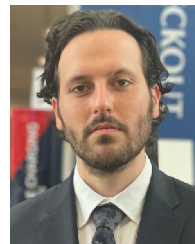
truss structure with simple support was used in the numerical validation of the presented approach. The ML algorithms are fed by applied external loadings with time operation as input and the real-time response at fourteen nodal locations as output. Based on the external loadings, ML algorithms predict real-time responses at all fourteen virtual sensors. The FE analysis is employed for training the ML models, which then predict the dynamical states of the truss structure at any time instant. The proposed surrogate model requires no re-run of supplementary FEA and significantly reduces the computational cost. Four ML algorithms including RF, XGBoost, Adaboost regression DTs, and DNNs have been used in a two-dimensional framework. The last contribution of this research was an extensive performance evaluation of the proposed approach. XGBoost regression outperforms other ML algorithms and provides more accurate results than the other three methods. However, all three DT algorithms have proven to be decent candidates that will serve as mid-fidelity surrogate models for FEA. We also discussed some limits of the proposed approach, including the fact that an ML model requires large training sets comprising millions of data records. The novel approach could employ larger and more complex networks (than those used in conventional approaches), but the computational cost increases for very large network sizes. Based on the results, we conclude that FEA can be effectively substituted with the proposed ML-based surrogate model to obtain the dynamic behavior of two-dimensional mechanical systems.

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